One More Trader Always Helps in Nested Symmetric Markets: A Note on Rostek and Weretka (2012)

A single entrant cannot change commonality, so information dilution needs entry in groups

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Abstract

Can one more trader make a price less informative? Rostek and Weretka (2012, *Econometrica*) show that in small markets it can, because entry can lower commonality, the average correlation between traders' values. Within their model, a trader added to a market without changing the correlations among incumbents cannot lower commonality. The equilibrium analysis requires every trader to have the same average correlation with the others, before and after entry, and this forces the entrant to correlate with each incumbent at exactly the existing commonality. Commonality is then unchanged and price informativeness rises with every entrant. The paper adopts this setting in Section 4.3 to explain its non-monotone examples, but neither the spatial nor the group example satisfies it. Information dilution in the model requires traders to arrive in groups or incumbents' correlations to change. We also correct Lemma 1, whose common value component need not exist under the stated condition, Proposition 4, whose criterion fails when commonality changes sign, and an inverted constant in Proposition 5. The paper's central possibility, that larger markets can have less informative prices, survives.

1 One trader at a time

Rostek and Weretka (2012), hereafter RW, show that a larger market can have a less informative price. The natural reading is that a newcomer can harm price discovery. In RW's model this reading fails. A trader added to a market whose incumbents keep their correlations must correlate with each of them at exactly the existing commonality, if the market is to remain symmetric enough for RW's equilibrium. Commonality therefore cannot move, and the price becomes more informative with every entrant (Proposition 1 and Corollary 2).

The argument uses only row sums of the correlation matrix. It applies directly to Section 4.3 of RW, which assumes this kind of nested entry in order to explain why informativeness can fall with market size, so within that section's assumptions informativeness cannot fall. We read Section 4.3 within RW's general framework, in which every finite auction in a sequence is equicommonal, because the symmetric equilibrium and the informativeness formula (1) are defined only for such auctions. Section 4.3 itself states only the limiting condition explicitly. Proposition 1 does not cover sequences whose finite markets are not equicommonal, and the paper's equilibrium and informativeness formula do not cover those markets either. RW's spatial model, in which commonality declines with size, cannot be generated by nested single entry, and the group model grows by two traders at a time. Entry in groups can dilute information with all correlations preserved. In an example with pairs of correlated traders, moving from two pairs to three lowers informativeness from $\frac{1}{55}$ to $\frac{1}{95}$.

A reader may object that RW never claim nested entry for their main results, and the objection is correct. They allow the correlation matrix to change arbitrarily with market size (p. 690), and their finite-market comparisons in Sections 2 to 4.2 stand. What the result rules out is reading those comparisons as the effect of one more trader joining an unchanged market, and the explanation offered in Section 4.3. Section 4 shows that Lemma 1 needs the average value to converge, and Section 5 corrects Propositions 4 and 5.

2 The setting

In Rostek and Weretka (2012), hereafter RW, I traders in a uniform-price double auction have jointly normal values θ_i with common variance and correlation matrix C, and each observes $s_i = \theta_i + \varepsilon_i$ with independent noise of relative variance σ^2. The matrix is *equicommonal* when every row has the same

off-diagonal sum, $C\mathbf{1} = [1 + (I-1)\bar{\rho}]\mathbf{1}$, and $\bar{\rho}$ is the commonality. Market size is $\gamma = 1 - 1/(I-1)$. In the symmetric linear equilibrium, price informativeness (RW's equation (20), p. 707) is

$$\psi(\gamma, \bar{\rho}) = \frac{\sigma^2 \bar{\rho}^2}{[(1+\sigma^2)(1-\gamma) + \bar{\rho}]\,[(1+\sigma^2) - \bar{\rho}]}. \tag{1}$$

At fixed $\bar{\rho} \neq 0$, ψ rises with γ, because market size enters only through the first factor of the denominator, which falls as γ rises. Growth can therefore lower informativeness only by lowering $|\bar{\rho}|$. In RW's examples the commonality function $\bar{\rho}(\gamma)$ varies with size, and in the spatial model it declines (p. 692).

RW allow the correlation matrix to change "in an arbitrary way" with market size (p. 690). Section 4.3, which decomposes signals into a common component and residuals to explain why informativeness can be non-monotone, makes a stronger assumption (p. 700): auction A_{I+1} results from adding a trader to A_I with the pairwise correlations of A_I preserved, so each finite auction is a truncation of one infinite vector of values.

3 A single entrant cannot change commonality

Proposition 1. *Let C_I be equicommonal with commonality $\bar{\rho}_I$, and let C_{I+1} add one trader and keep C_I as its leading $I \times I$ block. If C_{I+1} is equicommonal, then the entrant's correlation with every incumbent equals $\bar{\rho}_I$, and $\bar{\rho}_{I+1} = \bar{\rho}_I$.*

Proof. Let x_i be the correlation of incumbent i with the entrant. Incumbent i's off-diagonal row sum in C_{I+1} is $(I-1)\bar{\rho}_I + x_i$, which must equal $I\bar{\rho}_{I+1}$, so all x_i equal $x = I\bar{\rho}_{I+1} - (I-1)\bar{\rho}_I$. The entrant's row sum is Ix, which must also equal $I\bar{\rho}_{I+1}$. Hence $x = \bar{\rho}_{I+1}$ and $(I-1)(\bar{\rho}_{I+1} - \bar{\rho}_I) = 0$. □

Corollary 2. *In any infinite sequence of equicommonal auctions built by adding one trader at a time with incumbents' correlations preserved, commonality is a constant $\bar{\rho} \geq 0$, and price informativeness strictly increases with every entrant (if $\bar{\rho} > 0$) or is zero throughout (if $\bar{\rho} = 0$), at every size at which the symmetric equilibrium exists. It converges to $\sigma^2\bar{\rho}/(1+\sigma^2-\bar{\rho})$.*

Proof. Constancy follows from Proposition 1 by induction. The variance of the average value, proportional to $\bar{\rho} + (1-\bar{\rho})/I$, must stay nonnegative as I grows, so $\bar{\rho} \geq 0$. Monotonicity and the limit follow from (1). □

Because the argument uses only row sums, the starting market can have any correlation structure, and every later entrant correlates equally with all traders already present. In Section 4.3's setting the only force at work is therefore the direct effect of market size, which raises informativeness, and the non-monotonicity that the section sets out to explain cannot arise.

RW's examples do not fit this setting. The spatial model has decreasing commonality, so by Proposition 1 it cannot be built by single nested entry. The group model, whose U-shaped informativeness RW interpret with the Section 4.3 decomposition (p. 703), is defined only for even I and grows by adding one trader to each of two groups. Entry in groups does allow dilution with nested correlations. Take independent pairs of traders, with correlation $\frac{1}{2}$ within a pair and zero across pairs, and $\sigma^2 = 1$. Markets of two and three pairs are positive definite, equicommonal and nested, with commonality $\frac{1}{6}$ and $\frac{1}{10}$, and informativeness falls from $\frac{1}{55}$ to $\frac{1}{95}$. Both symmetric equilibria exist. The five-trader market in between is not equicommonal.

In RW's model, then, a newcomer who joins an existing market cannot dilute the information in prices. Dilution needs a larger market whose incumbents are correlated differently from those of the smaller one, or a group of newcomers who are highly correlated with one another and less so with incumbents. Proposition 4 should be read in the same way. Its change in commonality from "including a new bidder" (p. 697) is zero whenever the new bidder joins the old market and both markets are equicommonal.

4 Lemma 1 needs the average value to converge

RW's Lemma 1 (p. 709) states that a common value component exists, and equals $X = \lim_I I^{-1} \sum_{i \leq I} \theta_i$, if and only if the infinite auction is equicommonal, defined (p. 701) by

$$\lim_{I \to \infty} \frac{1}{I-1} \sum_{j \leq I,\, j \neq i} \rho_{ij} = \bar{\rho}_\infty \quad \text{for every fixed } i. \tag{2}$$

Condition (2) does not make the average converge. Split traders into consecutive sectors of sizes $1, 2, 4, 8, \ldots$, and let $\theta_i = \sqrt{b}\, Z_k + \sqrt{1-b}\, U_i$ for trader i in sector k, with independent standard normal Z_k and U_i and

$b \in (0,1)$. Every finite correlation matrix is positive definite, and (2) holds with $\bar{\rho}_\infty = 0$ because each trader shares a sector with finitely many others. But if A_K is the average over the first K complete sectors, $\mathrm{Var}(A_{K+1} - A_K) \to b/3 > 0$, so the average does not converge in probability. Each new sector is as large as all earlier ones together and moves the average by a shock of its own.

A Hilbert-space argument gives the exact repair. Under (2), the averages converge weakly to a unique X in the closed linear span of the values, with $\mathrm{Cov}(\theta_i, X) = \mathrm{Var}(X) = \bar{\rho}_\infty$ in correlation units, and the mean-square distance between the average of I values and X equals the variance of that average minus $\bar{\rho}_\infty$. The average converges, and Lemma 1 holds, if and only if that variance tends to $\bar{\rho}_\infty$. Single nested entry with equicommonal truncations delivers this, since the variance is proportional to $\bar{\rho} + (1-\bar{\rho})/I$. The sector example is not equicommonal at finite sizes, so the gap concerns the lemma's statement, not Section 4.3's setting.

5 Two smaller points

Proposition 4 across a sign change. Proposition 4 (p. 697) fixes γ and $\bar{\rho} > 0$ ($\bar{\rho} < 0$) and states that an entrant raises informativeness if and only if the change in commonality exceeds a threshold $\tau < 0$ (falls below $\tau > 0$). The proof (p. 708) uses only the root of (1) with the sign of $\bar{\rho}$, although it notes that each informativeness level has a positive and a negative root. When commonality can change sign, the correct criterion is that the new commonality lies outside the interval between the two roots. For example, with $\sigma^2 = 1$ and equicorrelated matrices, going from $I = 3$ with $\bar{\rho} = -\frac{1}{10}$ to $I = 4$ with $\bar{\rho} = \frac{1}{10}$ raises informativeness from $\frac{1}{189}$ to $\frac{3}{437}$, by 30%. Both matrices are positive definite and both equilibria exist. The change in commonality, $\frac{1}{5}$, is well above $\tau = 0.0197$, so the stated criterion predicts a fall.

The constant in Proposition 5. Below equation (9) (p. 701), $\bar{c}$ is defined as $\mathrm{Var}(s_i)\mathrm{Var}(p) - \mathrm{Cov}(s_i, p)^2$, the determinant of the covariance matrix of (s_i, p). Equation (9) needs its reciprocal, and the proof (p. 709) writes $\bar{c} \equiv 1/\det(\Sigma) = \mathrm{Var}(s_i)\mathrm{Var}(p) - \mathrm{Cov}(s_i, p)^2$, which contradicts itself. The levels of the inference coefficients change, but not their signs, which are what the paper uses.

6 Why it matters

RW's paper is a standard reference for the claim that a larger market need not have a more informative price. Applications that read the claim as a statement about participation, for example about admitting new participants to a market one at a time, need more than the model delivers. In the model, a single trader added to a symmetric market without disturbing incumbents' correlations always makes the price more informative. Information is diluted only by entry that changes the correlation structure, such as a new group of traders who are highly correlated with one another and weakly correlated with incumbents, or a reorganization of the market as it grows. The finite-market comparisons in RW's Sections 2 to 4.2, where the correlation matrix may change freely with size, are unaffected.

Notes. Page numbers refer to the published article. All numbers were checked in exact rational arithmetic; equilibrium existence uses RW's Proposition 1 with their inference coefficient (6). We found no published correction, but the search was not exhaustive.